

Solution to system of equations

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September 29, 2026

In the following question, I recommend that you solve using iterative and direct methods (Gauss Elimination, LU decomposition and matrix inversion). Assume equilibrium condition ($V_{app} = 0$)

1. Write a C code for the solving the following differential equation

$$\frac{d}{dx}\epsilon\frac{d}{dx}V = -q(N_D(x) - N_A(x)) \quad (1)$$

(a) $N_D = 10^{26}, N_A = 10^{25}$

(b) $N_A = 10^{26}, N_D = 10^{25}$

2. Solve the non-linear version of the Poisson equation as

$$\frac{d}{dx}\epsilon\frac{d}{dx}V = -q(N_D(x) - N_A(x) + p_0\exp(-qV/k_BT) - n_0\exp(qV/k_BT)) \quad (2)$$

for the following cases

- n-type semiconductor ($N_D = 10^{24}cm^{-3}$)
- p-type semiconductor ($N_A = 10^{24}cm^{-3}$)
- n-type (for $-1\mu m < x \leq 0$) and p-type (for $1\mu m > x > 0$)

The third case is a diode. To verify you code you can compare your answers with the analytical formulae you have studied in SDF